

Def. Let  $R$  be a Commutative Ring with  $1 \neq 0$ .

The polynomial ring in the indeterminate  $x$  with coefficients from  $R$  is:

$$R[x] \stackrel{\text{def}}{=} \bigcup_{n=0}^{\infty} \prod_{k=0}^{\infty} R_{k,n} \quad \text{where} \quad R_{k,n} = \begin{cases} R & \text{if } k \leq n \\ \{0\} & \text{if } k > n. \end{cases}$$

Denote a tuple  $(a_0, a_1, \dots, a_n, 0, 0, \dots) \in \prod_{k=0}^{\infty} R_{k,n}$  as:

$$a_0 + a_1x + \dots + a_nx^n$$

And, the addition is given by entry-wise, the multiplication is given by

$$\left( \sum_{i=0}^n a_i x^i \right) \cdot \left( \sum_{j=0}^m b_j x^j \right) = \sum_{k=0}^{n+m} \underbrace{\left( \sum_{i=0}^k a_i b_{k-i} \right)}_{\text{Cauchy product}} x^k$$

Prop. Let  $I \subset R$  be an Ideal of  $R$ . Then, in  $R[x]$ ,

$$(I) = I[x]$$

Moreover  $R[x]/(I) = R[x]/I[x] \cong (R/I)[x]$ .

Proof.  $(I) = I[R[x]] = (IR)[x] = I[x]$ , since  $R$  is commutative, and has 1.

Now, define a map  $\varphi: R[x] \rightarrow (R/I)[x]: \sum a_n x^n \mapsto \sum (a_n + I)x^n$

Then,  $\ker \varphi = I[x]$ , so proved by the first isomorphism theorem.

(More specifically,

$$\sum a_n x^n \in \ker \varphi \Leftrightarrow \sum (a_n + I)x^n = 0$$

$$\Leftrightarrow \text{all coefficients } a_n \text{ are contained in } I$$

$$\Leftrightarrow \sum a_n x^n \in I[x].$$



Thm. If  $R$  is Noetherian Ring, then  $R[x]$  is Noetherian.

Proof Let an ideal  $I \subseteq R[x]$  be given, and let  $L \subseteq R$  be the set of all leading coefficients of the elements in  $I$ . (Claim  $L$  is ideal of  $R$ ).

Since  $0 \in I$ , so  $0 \in L$ . Let  $a, b \in L$  and  $r \in R$  be given.

Then, there are two polynomials in  $I$ ,

$$f(x) = ax^d + \dots \quad \text{and} \quad g(x) = bx^e + \dots$$

Since  $I$  is Ideal,

$$r \cdot x^e \cdot f(x) - x^d \cdot g(x) \text{ is contained in } I, \text{ so } ra - b \in L.$$

Which shows  $L \subseteq R$  is an Ideal. Now,  $L$  is finitely generated by assumption for  $R$ .

Denote  $L = (a_1, \dots, a_n)$  where  $a_1, \dots, a_n \in R$ .

And, pick  $f_i \in I$  be a polynomial of degree  $e_i$  with the leading coefficient is  $a_i$ .

Let  $N = \max\{e_1, \dots, e_n\}$ . For each  $d \in \{0, 1, \dots, N-1\}$ , construct

$L_d$  be the set of all leading coefficients of polynomial in  $I$  of degree  $d$ , with  $0$ .

Similarly, each  $L_d$  is Ideal of  $R$ , so these are generated finitely:

$$L_d = (b_{d,1}, \dots, b_{d,n_d}) \text{ for some } b_{d,i} \in R.$$

Now, for  $f_{d,i}$ , the polynomial of degree  $d$  with the leading coefficient is  $b_{d,i}$ ,

the set  $\{f_1, \dots, f_n\} \cup \{f_{d,i} \mid 0 \leq d < N, 1 \leq i \leq n_d\}$  generates  $I$ , because:

Clearly the generated ideal is contained in  $I$ . If there is a polynomial  $f(x) \in I$  which is not contained in the generated ideal with the minimal degree.

If  $\deg f \geq N$ , then the leading coefficient  $a$  of  $f$  can be represent as:

$$a = r_1 a_1 + \dots + r_n a_n \quad (r_i \in R), \text{ } R\text{-linear combinations of generators for } I.$$

Then,  $g = r_1 x^{d-e_1} f_1 + \dots + r_n x^{d-e_n} f_n$  has a degree smaller than  $\deg f$ , with the leading coefficient is  $a$ . Contradiction to the minimality.

If  $\deg f < N$ ,  $a \in L_d$  ( $d < N$ ), so contradiction.

□